25900544 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHL200 - SYMBOLIC LOGIC

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **8** questions

Each question carries **2** marks

1. When is an argument considered valid? [U] / [CO1]

2. What is the role of symbols in logic? [K] / [CO1]

3. Define a proposition [K] / [CO1]

4. If p is true and q is false determine the truth value of p . q [K] / [CO2]

5. Write an expression in English to satisfy the symbolisation p.q [U] / [CO2]

6. State the justification for each line in the following formal proof of validity [K] / [CO3]

   1. A. B
   2. (A V C)→D/ ∴A. D
   3. A
   4. A V C

5 D

6. A. D

7. Give the statement form for [K] / [CO3]
   X V (Y. Z)

8. Give two examples of contingent statements [K] / [CO3]

9. Define modus tollens [K] / [CO3]

10. Symbolise ' No child is a voter ' in predicate logic [K] / [CO4]

11. Symbolise ' Some flowers are red' in predicate logic [K] / [CO4]

12. State the auxiliary symbol for a particular proposition [K] / [CO4]

[2x8 = 16]

Part B

Short essay

Answer any **6** questions

Each question carries **5** marks

13. How does the nature of symbolic logic elevate its scope in reasoning? [K] / [CO1]

14. Distinguish between traditional and symbolic logic [K] / [CO1]

15. Write the truth table for material implication [K] / [CO2]

16. If A and B are true and X and Y are false, but the values of P and Q are unknown, what can be said about the truth values of the following statements. [U] / [CO2]
    a. P. ~P
    b. P V ~Y

17. Using truth tables determine the validity or invalidity of the following [U] / [CO3]
    If Alice is elected president, then Betty is elected Vice president or Carol is elected treasurer. Betty is elected Vice president. Therefore, if Alice is elected president, then Carol is not elected treasurer.

18. Write the specific form of the following [U] / [CO3]
    a. J→(K. L)
    J V (K. L)
    ∴K . L
    b. Y V (Z. ~Y)

Y

∴ ~ (Z. ~Y)

19. [U] / [CO3]

Using truth table method determine whether the following are tautologous

a. p→(q→~r)

b. ~(p V q) V ~r

20. Write a short note on statement form with examples [K] / [CO3]

21. Distinguish between simple and compound propositions with examples. [K] / [CO2]

22. Symbolise the following in predicate logic [K] / [CO4]

a. Radha is intelligent

b. New Delhi is the capital of India

c. Ravi is an engineer

d. Govind is intelligent and hardworking

e. Columbus is a traveller

[5x6 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **12** marks

23. Critically examine the limitations of propositional logic and explain how predicate logic overcomes them. [U] / [CO1]

24. Discuss how constants and variables are used in propositional logic and predicate logic. Illustrate [U] / [CO2]

25. Using truth table method determine whether the following are contradictions: [U] / [CO3]

a.(p V q)→[(~p →r). (~q.r)

b. [(p →q).(q→r)]→(~p V r

26. Discuss how categorical propositions are analysed in predicate calculus [U] / [CO4]

[12x2 = 24]